

SIDDAGANGA INSTITUTE OF TECHNOLOGY-TUMAKURU 572103**Department of Computer Science and Engineering****Mini Project Synopsis for the Academic Year 2026-27**

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Title of the project	:	PhishGuard: AI-Based Phishing Website Detection Using Machine Learning		
Problem Statement	:	Phishing websites are a major cybersecurity threat that trick users into revealing sensitive information such as usernames, passwords, banking details, and personal data. Traditional security methods often fail to detect newly created or modified phishing websites. Users may unknowingly access malicious websites that closely resemble legitimate websites. Manual identification of phishing websites requires technical knowledge and can be time-consuming. Therefore, there is a need for an intelligent and automated system to identify suspicious websites accurately. The proposed PhishGuard system uses Artificial Intelligence and Machine Learning techniques to analyze website features and detect phishing attempts. The system examines factors such as URL characteristics, domain information, webpage behaviour, and other relevant features. It classifies websites as legitimate or phishing based on trained machine learning models.		
Abstract	:	Phishing attacks are one of the most common cybersecurity threats, where attackers create fake websites that imitate legitimate websites to steal sensitive information such as usernames, passwords, banking details, and personal data. PhishGuard: AI-Based Phishing Website Detection Using Machine Learning is a mini project designed to detect and classify phishing websites automatically. The system analyzes various features of a website, including URL structure, domain information, webpage characteristics, and other relevant parameters. Machine learning algorithms are trained using phishing and legitimate website datasets to identify patterns associated with malicious websites. Based on these features, the system predicts whether a given website is legitimate or phishing. The proposed system aims to provide quick and accurate detection while reducing the risk of users accessing fraudulent websites.		

Objectives	:	1. To develop an AI-based system for detecting and identifying phishing websites automatically. 2. To analyze website and URL features that are commonly associated with phishing attacks. 3. To train and evaluate machine learning models using phishing and legitimate website datasets. 4. To classify websites accurately as either legitimate or phishing based on extracted features. 5. To reduce the risk of users accessing malicious websites and protect sensitive information from cyber threats. 6. To provide a simple and user-friendly phishing detection system that delivers fast and reliable results.
Applications	:	• Web Browser Security – Can be integrated into browsers to identify and warn users about phishing websites. • Online Banking Protection – Helps detect fraudulent banking websites and protect users' financial information. • E-Commerce Security – Identifies fake shopping websites designed to steal customer credentials and payment details. • Email & Message Security – Can analyze links received through emails, SMS, and social media to detect phishing URLs. • Educational Institutions – Helps students and staff identify malicious websites and improve cybersecurity awareness.
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